FINDING TIME COMPLEXITY OF ALGORITHM

Problem 1: Finding Complexity using Counter Method

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
    int s =1;
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

PROGRAM;

```
#include<stdio.h>

void function(int n){

  int count=0;

  int i=1;

  count++;

  int s=1;

  count++;
```

```
    while(s<=n){
        count++;
        i++;
        count++;
        s+=i;
        count++;
    }
    count++;
    printf("%d",count);
}
int main(){
    int n;
    scanf("%d",&n);
    function(n);
}
```

OUTPUT:

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Problem 2: Finding Complexity using Counter method

```
Convert the following algorithm into a program and find its time complexity using the
counter method.

void func(int n)
{
```

```
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

PROGRAM:

```
#include<stdio.h>

void func(int n)

{

  int count=0;

  if(n==1)

  {

    count++;

    count++;

  }

  else

  {

    count++;

    for(int i=1; i<=n; i++)

   {

    count++;

   for(int j=1; j<=n; j++)
```

```
        {
          count++;
          count++;
          count++;
          break;
          count++;
        }
       count++;
     }
     count++;
   }
   printf("%d",count);
 }
 int main(){
     int n;
     scanf("%d",&n);
     func(n);
 }
```

OUTPUT:

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Problem 3: Finding Complexity using Counter Method

Convert the following algorithm into a program and find its time complexity using counter method.

```
 Factor(num) {
 {
    for (i = 1; i <= num;++i)
    {
     if (num % i== 0)
        {
          printf("%d ", i);
        }
     }
  }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

PROGRAM:

```
#include<stdio.h>
void Factor( int num)
 {
    int count=0;
    for (int i = 1; i <= num;++i)
   {
      count++;
      if (num % i== 0)
      {
         count++;
      }
      count++;
    }
    count++;
    printf("%d",count);
```

```
 }
int main(){
   int num;
   scanf("%d",&num);
   Factor(num);
}
```

OUTPUT:

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Problem 4: Finding Complexity using Counter Method

QUESTION:

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

 A positive Integer n

Output:

Print the value of the counter variable

PROGRAM:

```
#include<stdio.h>
void function(int n)
{
    int count=0;
    int c= 0;
    count++;
    for(int i=n/2; i<n; i++)
    {
        count++;
        for(int j=1; j<n; j = 2 * j)
        {
            count++;
            for(int k=1; k<n; k = k * 2)
            {
                count++;
                c++;
                count++;
            }
            count++;
        }
        count++;
    }
    count++;
    printf("%d",count);

}
int main(){
    int n;
    scanf("%d",&n);
    function(n);
```

}

OUTPUT:

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Problem 5: Finding Complexity using counter method

QUESTION:

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

PROGRAM:

```
#include<stdio.h>
void reverse(int n)
{
   int count=0;
   int rev = 0, remainder;
   count++;
   while (n != 0)
   {
      count++;
      remainder = n % 10;
      count++;
      rev = rev * 10 + remainder;
      count++;
      n/= 10;
      count++;
   }
   count++;
   count++;
   printf("%d",count);
}
int main (){
   int n;
   scanf("%d",&n);
   reverse(n);
}
```

OUTPUT:

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓